

Cognate/Professional Electives

OPTICS

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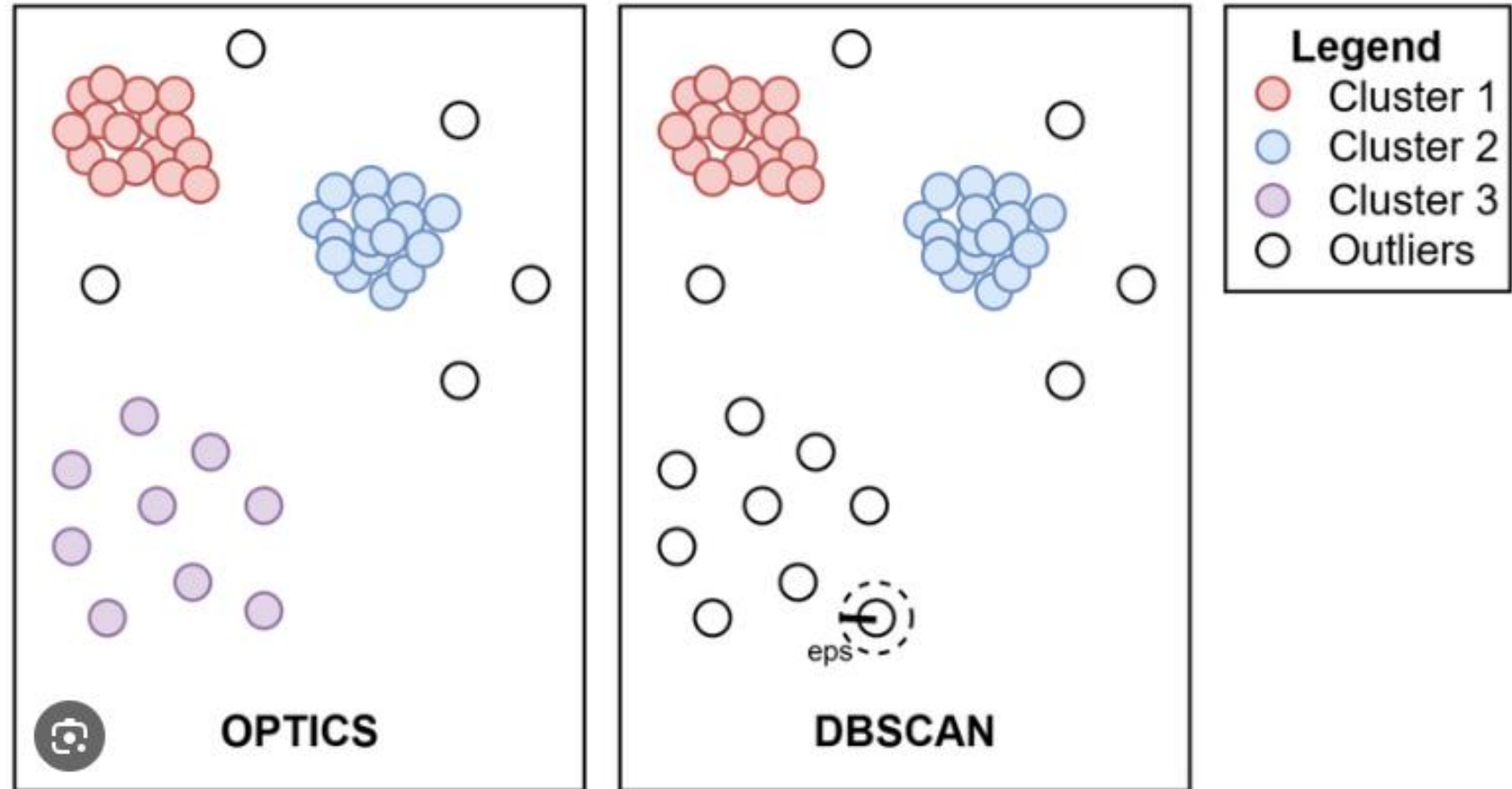
Ordering Points To Identify the Clustering Structure

- is a density-based clustering algorithm that is an **extension** of DBSCAN
- designed to address some of the **limitations** of DBSCAN by providing a richer and more robust view of the data's cluster structure

Core Concept

- it does not explicitly produce a clustering like DBSCAN, it creates an **augmented ordering** of database representing its density-based structure

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OPTICS is ideal for:

- Datasets with varying density clusters
- Understanding hierarchical density-based structures
- Applications where DBSCAN would struggle due to a single **global epsilon** value

Core Components:

- **Core Distance:** Minimum radius required to include at least minPts neighbors around a point
- **Reachability Distance:** Distance between two points **p** and **q** defined as:
- **Ordering:** OPTICS builds an ordering of data points based on their **reachability distances** which allows for easier visualization and cluster extraction

Key Differences Between OPTICS and Others

OPTICS vs. DBSCAN

OPTICS avoids the need for a global epsilon by building a reachability plot.

Better for datasets with **clusters of varying density**.

DBSCAN gives a hard clustering; OPTICS gives a **cluster ordering** that can be used flexibly.

OPTICS vs. KMeans

KMeans assumes spherical clusters and requires specifying k .

OPTICS detects arbitrary shapes and automatically infers the number of clusters.

KMeans is distance-based, while OPTICS is density-based.

OPTICS vs. Hierarchical Clustering

Hierarchical methods build a dendrogram but don't consider **density**.

OPTICS provides **density-aware hierarchical structures**.

Hierarchical methods are **greedy**, while OPTICS is more **global** in approach.

OPTICS vs. Gaussian Mixture Models

GMM assumes data is generated from a **mixture of Gaussians**.

OPTICS is **non-parametric** and makes **no assumptions** about distribution.

GMM fails with non-Gaussian, irregularly shaped clusters.

OPTICS vs. Spectral Clustering

Spectral clustering relies on the eigenvalues of a similarity matrix, which is computationally expensive.

Spectral clustering works well for complex boundaries, but doesn't handle density variation or noise well.

OPTICS is more **robust to noise and density fluctuations**.

Thank you very much for listening.